

**Internship Report**  
Of  
**Machine Learning Solutions for Agribusiness: Strategic Planning,  
Data Engineering, Model Development, and Deployment Strategy for  
Crop Disease Detection (Project AgriVision)**

Submitted  
To  
**School of Computer Science and Engineering  
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

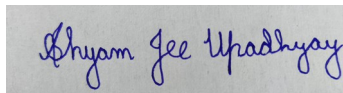
**B.Tech CSE**



By  
Roll Number of Student: 2410030380  
**Name of Student: SHYAM JEE UPADHYAY**  
Section: 3CSE3  
Batch: 2024-2028

## **CANDIDATE'S DECLARATION**

I, Shyam Jee Upadhyay, do hereby declare that the internship report titled “Machine Learning Solutions for Agribusiness: Strategic Planning, Data Engineering, Model Development, and Deployment Strategy for Crop Disease Detection (Project AgriVision)” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.



Signature

Student Name: Shyam Jee Upadhyay

Roll No.: 2410030380

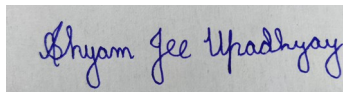
## ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Mr. Kounal Gupta, Founder of YuvaIntern, and the entire YuvaIntern and Henry Harvin Education team for the opportunity to work as a Junior Machine Learning Engineer - Agribusiness intern and for their mentorship throughout the four weeks of this internship.

I also express my gratitude to my faculty at the School of Computer Science and Engineering for their continuous support and guidance.

I express my gratitude to my parents for their blessings.



Signature

Student Name: Shyam Jee Upadhyay

Roll No.: 2410030380

# INTERNSHIP COMPLETION CERTIFICATE



## CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **Shyam Jee Upadhyay** successfully completed an internship with **YuvaIntern** in the role of **Junior Machine Learning Engineer - Agribusiness**.

Throughout the tenure of this internship, Shyam demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend Shyam's hard work and wish them continued success in all future endeavors.

**Certificate No:** YI/2026/208251/451548

**Date of Issue:** October 08, 2026

FOR AND ON BEHALF OF YUVAINTERN



**Kounal Gupta**

Founder, [YuvaIntern.com](https://yuvaintern.com)

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## **4. PROJECT DESCRIPTION**

### **4.1 Introduction**

This report documents the work completed during a four-week internship undertaken as a Junior Machine Learning Engineer - Agribusiness with YuvaIntern (in association with Henry Harvin Education), carried out remotely from September 10, 2026 to October 08, 2026. The internship was structured around a single applied machine learning initiative, internally referred to as Project AgriVision, which explores how machine learning can be used to support smallholder farmers through early, mobile-based crop disease detection.

Agriculture continues to employ a large share of the workforce in many developing regions, and a significant portion of that production comes from smallholder farms that rely on manual observation and have limited access to agronomists or diagnostic laboratories. Crop pests and disease remain one of the most persistent threats to farm productivity, and the delay between symptom onset and expert diagnosis is often the deciding factor between a manageable outbreak and a lost harvest. Advances in computer vision, transfer learning, and the growing penetration of smartphones in rural areas make it feasible to place an instant, reasonably accurate diagnostic tool directly into a farmer's hands, and this internship was designed to work through that opportunity in a structured, week-by-week manner covering strategic planning, data preparation, model development, and deployment.

### **4.2 Organization Profile**

YuvaIntern is a skill-development and internship platform that partners with Henry Harvin Education, a professional training and upskilling organisation, to place learners into short, project-based internships across multiple technology domains, including machine learning and data science. Internships delivered under this program are supported by the National Skill Development Corporation (NSDC) framework and are designed to give students hands-on, deliverable-driven experience in a real-world problem domain rather than purely theoretical exposure. This internship was completed under the Agribusiness track, focusing specifically on the application of machine learning to precision agriculture.

### **4.3 Problem Statement**

Smallholder farmers currently detect crop disease primarily through visual inspection and word-of-mouth, after which they must travel to, or wait for, an agricultural extension officer to confirm the diagnosis. This process is slow, inconsistent across regions, and does not scale with the number of farmers who need support during an

outbreak. By the time an official diagnosis and treatment plan arrive, the disease has often progressed beyond the point where low-cost intervention is effective, forcing farmers into either significant yield loss or unnecessary over-application of agrochemicals as a precaution. The core problem this project addresses is therefore how to close this diagnostic delay using an image-based machine learning model that a farmer can access directly from a smartphone, with a low-bandwidth fallback for areas with poor connectivity.

#### **4.4 Project Objectives**

The internship objectives were defined using a SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound) and scoped to what a pilot phase could realistically validate:

- Build an image classification model for the top five diseases affecting the pilot crop(s), targeting at least 90% top-1 validation accuracy across the target disease classes.
- Deliver a mobile/USSD interface for photo-based diagnosis that remains usable under intermittent connectivity, returning a diagnosis in under 15 seconds on a mid-range smartphone.
- Reduce the average time between symptom onset and confirmed diagnosis for pilot farmers from roughly 5-7 days to under 24 hours.
- Validate the real-world impact of the solution on yield loss and pesticide usage against a control group.
- Establish a data pipeline and feedback loop that supports continuous, automated model improvement beyond the initial pilot.

#### **4.5 Scope of the Project**

The project was scoped as an initial pilot covering one or two crop value chains, with a defined path to scale to additional crops and regions in later phases. The scope spanned the full lifecycle of an applied ML solution: strategic problem definition and roadmap planning (Week 1), data quality assessment and feature engineering for agricultural data sources (Week 2), model planning, algorithm selection and evaluation-metric design (Week 3), and deployment architecture together with a stakeholder communication strategy (Week 4). The target users considered throughout were smallholder farmers, agricultural extension officers, and agribusiness cooperatives, and the target datasets considered were crop yield records, weather data, soil quality measurements, market/price data, and remote-sensing (satellite) imagery.

#### **4.6 Technologies and Tools Used**

- Programming and ML libraries: Python, pandas, scikit-learn, XGBoost/LightGBM.
- Deep learning: Convolutional Neural Networks (CNNs) with transfer learning from pretrained backbones such as ResNet/EfficientNet, for image-based disease classification.
- Model explainability: SHAP values for tabular models, Grad-CAM overlays for CNN-based image models.
- Hyperparameter tuning: Grid Search, Random Search, and Bayesian Optimization (e.g., Optuna/scikit-optimize).
- Deployment infrastructure: containerized model serving (Docker) behind a managed cloud inference endpoint with autoscaling, a lightweight mobile/progressive web app with offline caching, and an SMS/USSD fallback channel.
- Data sources: public agricultural image datasets (e.g., PlantVillage-style labeled sets), IoT weather and soil sensors, satellite-derived vegetation indices (NDVI), and commodity market/price feeds.

#### **4.7 System Architecture**

The proposed AgriVision solution is organised into four layers. The data layer collects and curates labeled crop images from public agricultural datasets, partner cooperatives, and farmer-submitted photos. The model layer is built around a convolutional neural network or lightweight vision transformer, fine-tuned via transfer learning from an existing image backbone. The application layer consists of a mobile app with offline capture together with an SMS/USSD channel for feature-phone users. The insight layer aggregates anonymized diagnoses into a regional dashboard for extension officers and agribusiness partners.

For deployment, a hybrid cloud-edge architecture was recommended over a purely cloud-centric or purely on-device design: the trained model is containerized and served via a managed cloud endpoint with autoscaling, while field access is handled through a lightweight mobile/progressive web app with offline caching of the last known recommendation, and a low-connectivity fallback (SMS/USSD/WhatsApp-based query-response) reaches smallholder farmers without smartphones or reliable data plans. Data ingestion is handled through scheduled batch synchronisation from IoT sensors and manual field entries, backed by cloud data warehousing with a local on-device cache for last-mile resilience.

#### **4.8 Methodology**

The development approach combined domain-informed data preparation with a staged, auditable model-development workflow.

- Research and benchmarking: reviewing existing literature and open datasets to select an appropriate model architecture and establish an accuracy baseline.
- Data cleaning and feature engineering: distinguishing genuine agronomic variability (e.g., a heatwave or flood) from noise (sensor or entry errors), using time-aware interpolation for short sensor gaps, spatial (KNN) imputation for missing soil tests, per-crop/per-region outlier thresholds, and engineered features such as Growing Degree Days, rainfall distribution features, nutrient ratios, and a soil fertility composite score.
- Model planning and training: a chronological, group-aware train/validation/test split to prevent data leakage across seasons or farms, a simple baseline model to set a minimum performance bar, and k-fold cross-validation for stable performance estimates.
- Algorithm selection: Gradient Boosting (XGBoost/LightGBM) as the primary engine for tabular yield and price problems, Random Forest as a robust fallback, and a CNN with transfer learning for image-based disease detection, selected against interpretability, speed, and scalability criteria relevant to agribusiness deployments.
- Human-in-the-loop validation: routing low-confidence predictions to a human agronomist for review, both to protect accuracy and to generate new labeled training data.
- Iterative field testing: staged deployment starting with a small group of cooperating farmers before expanding to the full pilot cohort.

#### **4.9 Expected Outcomes**

Success is assessed through a combination of technical and real-world outcome metrics, reviewed at the end of the pilot phase.

- Model quality: validation accuracy, precision/recall per disease class, and consistency across field-condition test images.
- System performance: on-device inference speed, app stability, and successful operation under low-connectivity conditions.
- Operational impact: reduction in diagnosis turnaround time and measurable decrease in disease-related yield loss versus a control group.
- Adoption: number of active farmers using the tool, repeat usage rate, and qualitative feedback from onboarding sessions.

- Business impact: reduced input waste, improved yield-forecasting accuracy for procurement planning, and a deployment architecture that scales to new regions without a proportional increase in cost.

Overall, the internship is expected to produce a clear, evidence-based recommendation on whether and how to scale the AgriVision solution to additional crops, regions, and farming communities in subsequent phases.

#### 4.10 Certificates of Completion and Other Communication Proof

The following pages contain the Internship Offer Letter and the Certificate of Experience issued by YuvaIntern (in association with Henry Harvin Education and NSDC) on completion of the internship.

### *Internship Offer Letter*



#### INTERNSHIP OFFER LETTER

Date of Issue: September 10, 2026

Dear Shyam Jee Upadhyay,

We are pleased to extend to you this formal offer for an Internship position as **Junior Machine Learning Engineer - Agribusiness** at **YuvaIntern**. We were impressed with your background, skills and enthusiasm, and we believe you will be a valuable addition to our team.

##### Internship Details

**Start Date** : September 10, 2026  
**End Date** : October 08, 2026  
**Duration** : 4 weeks  
**Location** : Remote

##### Key Responsibilities:

As a **Junior Machine Learning Engineer - Agribusiness** intern, you will work on tasks and projects related to As a Junior Machine Learning Engineer in the Agribusiness sector, you will be responsible for developing and implementing machine learning models to optimize agricultural processes and improve crop yield. You will work closely with agronomists and data scientists to analyze data, build predictive models, and provide insights for precision agriculture..

##### Benefits & Learning Opportunities:

Throughout the duration of your internship, you will have the opportunity to:

- Develop practical skills in [relevant skills/tools].
- Gain hands on experience in [specific tasks/projects].

As part of joining our team, you may be required to sign a confidentiality or non-disclosure agreement. You are also expected to comply with all applicable company policies and regulations, which will be provided to you upon commencement.

We look forward to welcoming you to the team.



**Kounal Gupta**  
 Founder, [YuvaIntern.com](https://YuvaIntern.com)

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## Certificate of Experience



### CERTIFICATE OF EXPERIENCE

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